Accelerating Large-Scale Decision Tree Induction on Heterogeneous CPU-GPU Architectures

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Abstract. We investigate parallel implementations of two widely used decision tree induction algorithms, CART and C4.5, on multicore CPUs, CUDA-enabled GPUs, and heterogeneous CPU-GPU architectures.

Keywords: Decision trees · CART · C4.5 · Parallel computing

1 Introduction

2 Related Work

2.1 Decision Tree Induction: CART and C4.5

[Brief: CART [2] (Gini, binary splits, cost-complexity pruning), C4.5 [5] (gain ratio, MDL threshold cost, error-based pruning). Only what is needed to read Sect. 3.]

2.2 Scalable and Parallel Induction on CPUs

2.3 Decision Trees on GPUs

3 Our Implementation and Optimizations

3.1 Simple

3.2 Serial

3.3 Parallel

3.4 GPU

3.5 Hybrid

4 Experiment Design

4.1 Datasets

[One sentence per dataset (domain, how the test set is defined). All features numeric; ECBDL's nominal features as integer codes. The GPUs train ECBDL on its first 15M rows and SOREL on its first 4M (about 9.5G values each, 73 GiB of the 80 GB VRAM).]

Table 1. Datasets. Sizes are of the training set as float32.

Dataset	Features	Classes	Training rows	Test rows	Size	Test set
SUSY [1]	18	2	4,500,000	500,000	0.3 GB	UCI split
HIGGS [1]	28	2	10,500,000	500,000	1.2 GB	UCI split
ECBDL'14 [6]	631	2	31,992,921	2,897,917	80.8 GB	official
GPU sample	631	2	15,000,000	2,897,917	37.9 GB	official
MNIST8M [4]	784	10	8,100,000	10,000	25.4 GB	MNIST test
SOREL-20M [3]	2381	2	12,699,013	4,195,042	120.9 GB	official
GPU sample	2381	2	4,000,000	4,195,042	38.1 GB	official

4.2 Hardware

[Four nodes of the mordor3 cluster (Table 2), each used whole (exclusive jobs). GPU nodes run the CUDA backend only, CPU nodes the CPU backends and the other CPU implementations. GPU runs are bound to the GPU's NUMA node (16 CPUs on hopper, 24 on ampere). Add measured memory bandwidth (STREAM) per node. Software: Ubuntu 24.04, Linux 7.0, g++ 13.3 (`-O3`), CUDA 12.9, NVIDIA driver 615.71.]

Table 2. Hardware. Every node has two sockets and no SMT; CPU, cores and L3 are per socket. One GPU (PCIe) is used per run.

Node	CPU (per socket)	Cores	Clock	L3	RAM	GPU
hopper	Xeon Gold 6444Y	16	4.0 GHz	45 MB	1 TB DDR5-4800	H100 80 GB
ampere	EPYC 7402	24	3.35 GHz	128 MB	1 TB DDR4-3200	A100 80 GB
epyc1	EPYC 9255	24	4.3 GHz	128 MB	512 GB DDR5-6400	–
dell1	Xeon Gold 6542Y	24	4.1 GHz	60 MB	1 TB DDR5-5200	–

4.3 Compared Implementations

[./tree backends (`-simple`, `-serial`, `-parallel`, `-cuda`, `-gpu-only`); scikit-learn and Weka J48 on the CPU; cuML and XGBoost on the GPU, with their settings. Why each is the closest equivalent (no public GPU C4.5; J48 depth limit adapter; XGBoost squared error = Gini on bins). rpart and YaDT: add or say why not.]

4.4 Protocol and Measurement

[One tree, depth $\leq$ 10, no pruning, all rows and features; what is timed (training only, from data in memory; GPU setup and upload included); peak RAM and VRAM; runs per case, means and spread; exclusive nodes; fairness (CPU threads of the hybrid, warm library processes, input layout). Source: docs/GPU-LIBRARIES.md, bench/README.md.]

5 Results

5.1 Training Time

[Overview of every implementation on every dataset (Tables 3 and 4): speed-ups of Serial-Opt over Serial-Simple, Parallel over Serial-Opt, GPU over Parallel, and of this work over the existing implementations. Variance: report mean with min–max, or mean ± 95 % confidence interval, once every cell has 10 runs.]

Table 3. Training time of one CART tree (depth ≤ 10, no pruning), in seconds: mean, and below it the fastest–slowest run. † marks a rough estimate (no run yet) with a plausible range below it; – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Parallel 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs.

This work Dataset	Serial-Simple	Serial-Opt	Parallel	GPU	Hybrid
SUSY	70.8 70.8–70.8	2.84 2.81–2.87	0.239 0.215–0.258	0.082 0.072–0.084	0.084 0.077–0.093
HIGGS	236 236–237	10.4 10.4–10.5	0.607 0.579–0.633	0.150 0.147–0.153	0.151 0.145–0.156
ECBDL'14	5 860	503 500–506	13.5 13.5–13.6	–	–
15M sample	2 270	225 225–225	6.32 6.29–6.34	2.97 2.96–2.98	2.97 2.96–2.98
MNIST8M	1 880	234 233–235	5.92 5.90–5.94	3.57 3.54–3.57	3.56 3.54–3.57
SOREL-20M	14 100	878	21.8 21.2–22.2	–	–
4M sample	*4 000*† 2 500–6 000	*280*† 200–400	*7*† 5–10	*3.5*† 2.5–6	*3.5*† 2.5–6
Existing implementations Dataset	scikit-learn	rpart		cuML	XGBoost
SUSY	86.5 86.2–87.1	*140*† 90–250		0.133 0.126–0.147	1.43 1.42–1.45
HIGGS	215 214–215	*350*† 200–600		0.456 0.456–0.456	3.43 3.29–3.58
ECBDL'14	*4 500*† 3 000–9 000	*7 500*† 4 000–15 000		–	–
15M sample	2 200	*3 600*† 2 000–6 000		9.95 9.90–10.0	42.2 42.1–42.4
MNIST8M	*2 500*† 1 500–5 000	*4 000*† 2 000–8 000		*7*† 4–14	*50*† 25–100
SOREL-20M	*13 000*† 7 000–26 000	*21 000*† 10 000–50 000		–	–
4M sample	*3 500*† 2 000–7 000	*6 000*† 3 000–15 000		*10*† 5–20	*45*† 25–90

5.2 Comparison with Other CPU Implementations

[./tree's backends against scikit-learn (CART) and Weka J48 (C4.5): time, memory, test accuracy, same trees.]

Table 4. Training time of one C4.5 tree (depth ≤ 10, no pruning), in seconds: mean, and below it the fastest–slowest run. † marks a rough estimate (no run yet) with a plausible range below it; – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Parallel 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs. cuML grows an entropy tree, its closest equivalent.

This work					
Dataset	Serial-Simple	Serial-Opt	Parallel	GPU	Hybrid
SUSY	77.2 77.1–77.2	3.10 3.06–3.12	0.249 0.221–0.308	0.109 0.094–0.114	0.112 0.108–0.116
HIGGS	257 257–257	11.9 11.9–12.0	0.590 0.568–0.628	0.164 0.159–0.171	0.170 0.167–0.172
ECBDL'14	6 890	623	17.6 17.5–17.7	–	–
15M sample	2 590	277 277–277	7.81 7.76–7.86	2.99 2.99–3.00	2.99 2.99–3.00
MNIST8M	1 970	209 209–209	5.64 5.60–5.67	3.71 3.69–3.72	3.68 3.66–3.69
SOREL-20M	14 200	1 130	28.8 28.6–29.0	–	–
4M sample	*4 000*† 2 500–6 000	*360*† 250–500	*9*† 6–13	*3.5*† 2.5–6	*3.5*† 2.5–6

Existing implementations			
Dataset	Weka J48	YaDT	cuML
SUSY	230 229–230	*46*† 25–80	0.128 0.119–0.139
HIGGS	830	*170*† 90–300	0.433 0.414–0.457
ECBDL'14	*50 000*† 25 000–100 000	*10 000*† 5 000–20 000	–
15M sample	24 200	*4 800*† 2 500–9 000	9.89 9.84–9.98
MNIST8M	*14 000*† 7 000–28 000	*2 700*† 1 200–6 500	*7*† 3–13
SOREL-20M	*60 000*† 30 000–120 000	*12 000*† 5 000–30 000	–
4M sample	*18 000*† 9 000–40 000	*3 800*† 1 500–9 000	*10*† 5–20

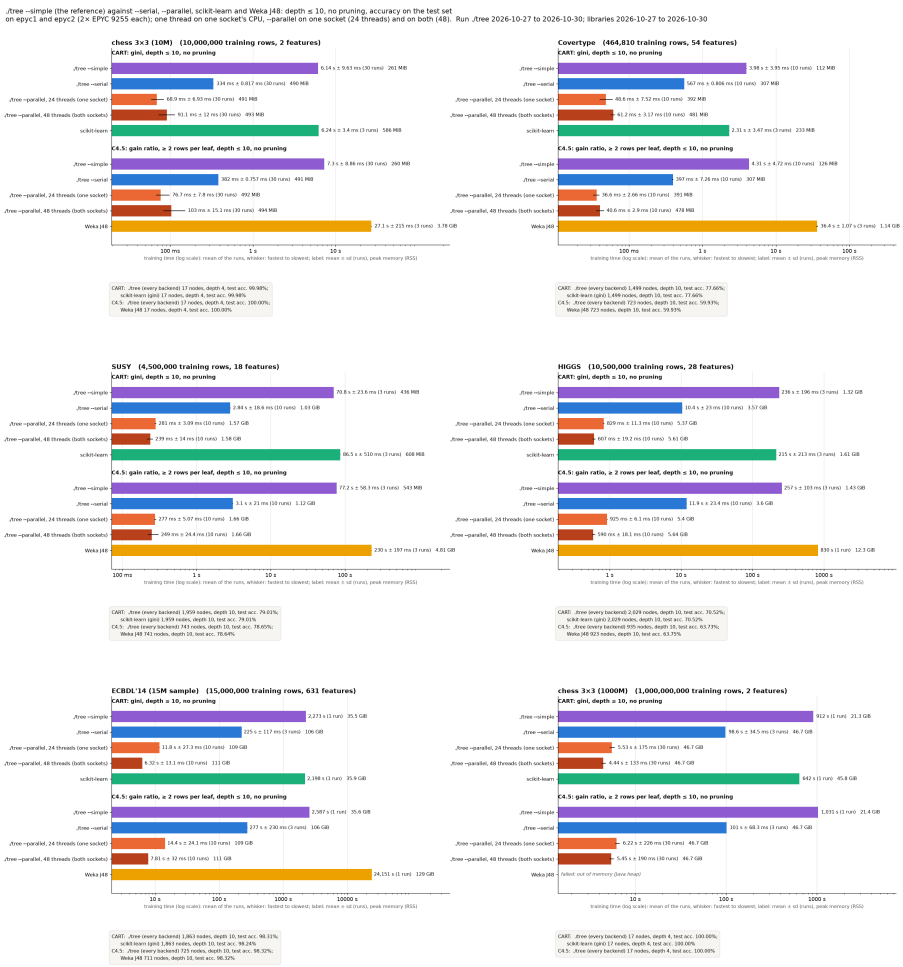


Fig. 1. Temporary: ./tree against scikit-learn and Weka J48 (cluster-cpu-vs-others).

5.3 Comparison with GPU Libraries

[Against cuML and XGBoost: one tree, and per tree of a 10-tree forest; accuracy of exact versus binned trees; VRAM.]

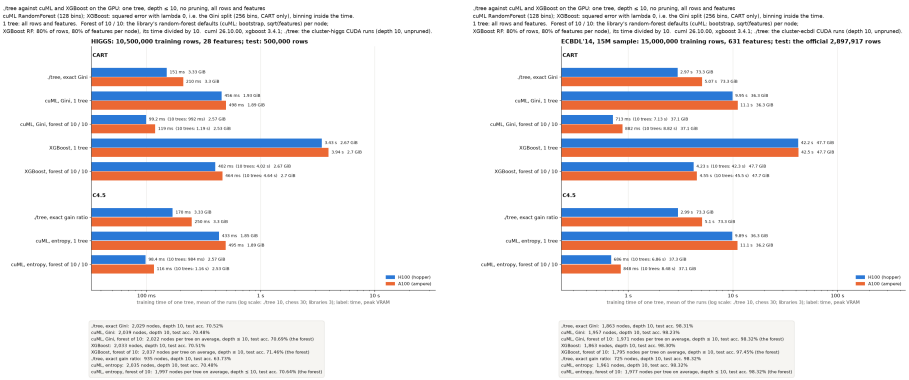


Fig. 2. Temporary: ./tree against cuML and XGBoost on HIGGS (left) and ECBDL'14 15M (right) (cluster-gpu-vs-others).

5.4 Performance Across Hardware

[All ./tree backends on the four nodes, per dataset: serial, one and two sockets, H100 and A100 hybrid and GPU only. CPU against GPU; which hardware wins for which data shape (rows, features, classes).]

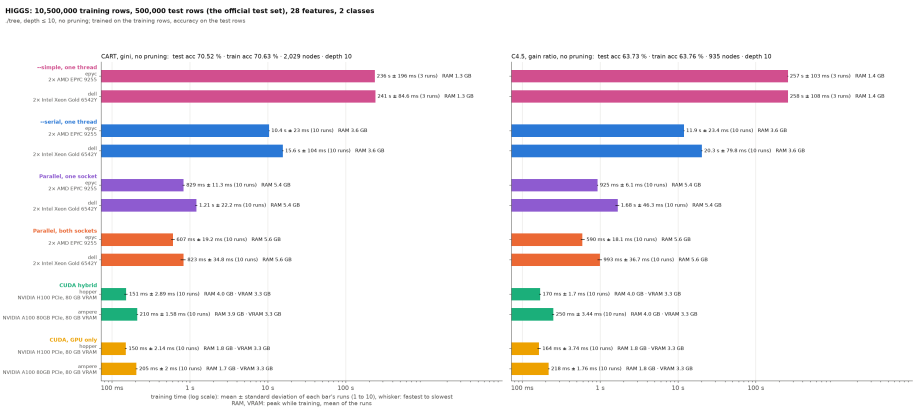


Fig. 3. Temporary: ./tree's backends on HIGGS on every node (cluster-higgs).

5.5 Scalability

[Thread scaling 1–48 on epyc1 and dell1 (memory bandwidth bound); growth with the number of rows (subsamples of one dataset).]

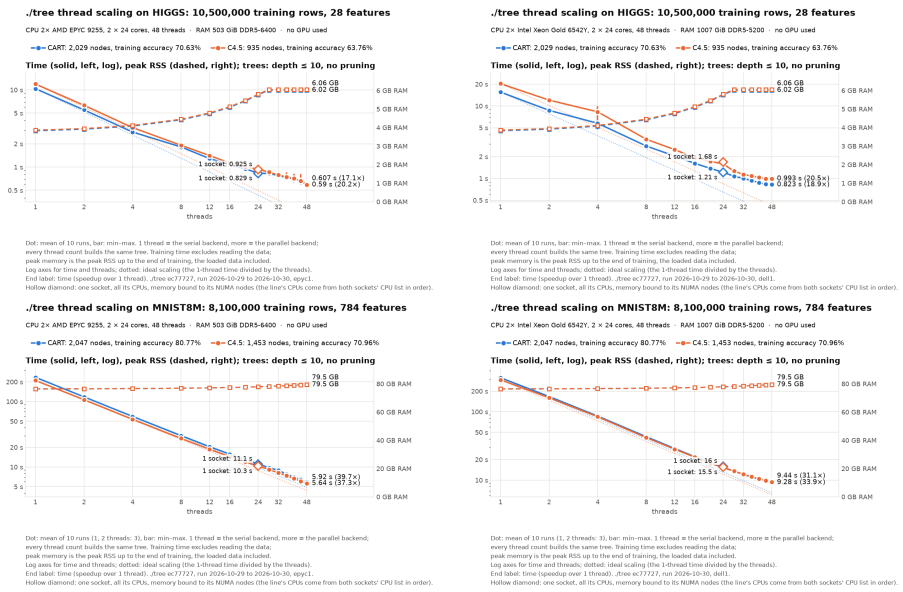


Fig. 4. Temporary: thread scaling on HIGGS and MNIST8M (cluster-cpu-scaling).

5.6 Profiling

[Where the time goes: presort, split search, partition, CPU–GPU transfer; GPU share of the hybrid; achieved bandwidth against peak.]

6 Discussion

7 Conclusion

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Fig. 5. Temporary: breakdown of the growing time (cluster-profiling).

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